17. Disinfectants An experiment was conducted to study the use of 95% ethanol versus 20% bleach as a disinfectant in removing contamination when culturing plant tissues. The experiment was repeated 15 times with each disinfectant, using cuttings of eggplant tissue.[9] The observation reported was the number of uncontaminated eggplant cuttings after a 4-week storage.

Disinfectant	95% Ethanol	20% Bleach
Mean	3.73	4.80
Variance	2.78095	.17143
n	15	15
	Pooled variance 1.47619	

a. Are you willing to assume that the underlying variances are equal?

b. Using the information from part a, is there evidence of a significant difference in the mean numbers of uncontaminated eggplants for the two disinfectants tested?

a. $F = \frac{s_1^2}{s_2^2} = \frac{2.78095}{0.17143} = 16.22 > 3$

$\therefore$ No

b. $H_0: \mu_1 = \mu_2$, $H_a: \mu_1 \neq \mu_2$

$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2(\frac{1}{n_1}+\frac{1}{n_2})}} \approx \frac{-1.07}{0.4436} \approx -2.41$

$df = 15+15-2 = 28$, $t_{28, 0.025} \approx 2.048$

$\because |-2.41| > 2.048$ $\therefore$ reject H_0

20. Dissolved O_2 Content A pollution control inspector suspected that a river community was releasing amounts of semitreated sewage into a river. To check his theory, he drew five randomly selected specimens of river water at a location above the town, and another five below. The dissolved oxygen readings (in parts per million) are as follows:

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

a. Do the data provide sufficient evidence to indicate that the mean oxygen content below the town is less than the mean oxygen content above? Test using $\alpha = .05$.

b. Estimate the difference in the mean dissolved oxygen contents for locations above and below the town. Use a 95% confidence interval.

a. $\bar{x}_1 = 5$, $s_1^2 = 0.025$

$\bar{x}_2 = 4.86$, $s_2^2 = 0.013$

$s_p^2 = \frac{0.025+0.013}{2} = 0.019$

$H_0: \mu_1 = \mu_2$, $H_a: \mu_1 > \mu_2$

$t = \frac{5-4.86}{\sqrt{0.019(\frac{1}{5}+\frac{1}{5})}} = \frac{0.14}{0.087} \approx 1.609$

$t_{8, 0.05} = 1.86$

$\because 1.609 < 1.86$ $\therefore$ non-reject H_0

b.

$(\bar{x}_1 - \bar{x}_2) \pm t_{8, 0.05}\sqrt{s_p^2(\frac{1}{5}+\frac{1}{5})}$

$= 0.14 \pm 2.306 \times 0.087 = 0.14 \pm 0.201$

$= (-0.061, 0.341)$ #

Paired-Difference Tests II *Use the information in Exercise 9 to calculate the observed value of the t statistic for testing the difference between the two population means using paired data. Find a rejection region with $\alpha = .05$ to state your conclusions. Construct a 95% confidence interval for $\mu_1 - \mu_2$. Does this confirm the results of your hypothesis test?*

9.

	Pairs				
Population	1	2	3	4	5
1	1.3	1.6	1.1	1.4	1.7
2	1.2	1.5	1.1	1.2	1.8

$d(x_1 - x_2) = \{0.1, 0.1, 0, 0.2, -0.1\}$

$\bar{d} = 0.06, \quad S_d = 0.114$

$H_0: \mu_1 - \mu_2 = 0, \quad H_a: \mu_1 - \mu_2 \neq 0$

$T = \frac{\bar{d} - \mu_d}{S_d/\sqrt{n}} = \frac{0.06}{0.114/\sqrt{5}} = 1.176$

$t_{4, 0.05} = 2.776$

R.R.: $|T| > 2.776$

∴ Non-reject H_0

95% C.I.

$0.06 \pm 2.776 \times 0.051 = 0.06 \pm 0.141$

$(-0.081, 0.201)$

DATA SET DS1026

20. Memory Experiments Twenty students participated in an experiment in which they were first asked to recall a list of 25 words. They were then given a second list, and asked to form images of the words as they were read. The number of words recalled for each student is given in the table.

Student	With Imagery (x_1)	Without Imagery (x_2)	$d(x_1 - x_2)$
1	20	5	15
2	24	9	15
3	20	5	15
4	18	9	9
5	22	6	16
6	19	11	8
7	20	8	12
8	19	11	8
9	17	7	10
10	21	9	12
11	17	8	9
12	20	16	4
13	20	10	10
14	16	12	4
15	24	7	17
16	22	9	13
17	25	21	4
18	21	14	7
19	19	12	7
20	23	13	10

Does it appear that the average recall score is higher when imagery is used?

$\bar{d} = \frac{205}{20} = 10.25$

$S_d = 4.115$

$t = \frac{\bar{d} - 0}{S_d/\sqrt{n}} = \frac{10.25}{4.115/\sqrt{20}} = \frac{10.25}{0.92} \approx 11.14$

$t_{19, 0.05} = 1.729$

$11.14 > 1.729$

∴ 使用 鼓勵商能不使用

DATA SET DS1028

22. Cake Mixes An experiment was conducted to compare the densities (in ounces per cubic inch) of cakes prepared from two different cake mixes. Six cake pans were filled with batter A, and six were filled with batter B. Expecting a variation in oven temperature, the experimenter placed a pan filled with batter A and another with batter B *side by side* at six different locations in the oven. The six paired observations of densities are as follows:

Location	1	2	3	4	5	6
Batter A	.135	.102	.098	.141	.131	.144
Batter B	.129	.120	.112	.152	.135	.163
d	.006	-.018	-.014	-.011	-.004	-.019

a. Do the data present sufficient evidence to indicate a difference between the average densities of cakes prepared using the two types of batter?

b. Construct a 95% confidence interval for the difference between the average densities for the two mixes.

$\bar{d} = -0.01$

$S_d = 0.00938$

(b) $\bar{d} \pm t_{df,\alpha/2} \times \left(\frac{S_d}{\sqrt{n}}\right)$

$= (-0.01985, -0.00157)$

(a) $H_0: \mu_d = 0$, $H_a: \mu_d \neq 0$

$t = \frac{\bar{d} - 0}{S_d/\sqrt{n}} = -2.61$

$t_{5,0.025} = 2.571$

$\because |-2.61| > 2.571$ $\therefore$ reject H_0

DATA SET DS1031

12. A Precise Quarterback The number of passes completed and the total number of passing yards recorded for the Los Angeles Chargers quarterback, Philip Rivers for each of the 15 regular season games that he played in the fall of 2017[13] were used to calculate the average number of yards per pass for each game.

Week	Completions	Yards per Pass	Week	Completions	Yards per Pass
1	28	13.8	9	17	12.5
2	22	13.2	11	15	12.2
3	20	11.4	12	25	10.7
4	18	17.7	13	21	12.3
5	31	11.1	14	22	15.8
6	27	16.1	15	20	11.8
7	20	12.6	16	31	10.7
8	21	17.2	17	22	8.7

a. Calculate the mean and standard deviation of the number of completions and the number of yards per pass.

b. Find a 95% confidence interval estimate for the variance of the number of completions. Why would you prefer a small variability in the number of completed passes?

c. Find a 95% confidence interval for σ^2, the variance of the yards per pass. Use these results to find a 95% confidence interval for σ, the standard deviation of the yards per pass.

d. Test whether the standard deviation of the yards per pass for this quarterback differs from $\sigma = 4$ with $\alpha = .05$.

a. 計算平均值與標準差

- **完成次數:**
  - 平均值 $\bar{x} \approx 21.93$
  - 樣本標準差 $s_x \approx 4.38$
- **每傳碼數:**
  - 平均值 $\bar{y} \approx 13.125$
  - 樣本標準差 $s_y \approx 2.45$

b. 完成次數變異數的 95% 置信區間

公式：$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}\right]$

- $df = 14$，查表 $\chi^2_{.025,14} = 26.119$，$\chi^2_{.975,14} = 5.629$
- 計算：$\left[\frac{14 \times 19.21}{26.119}, \frac{14 \times 19.21}{5.629}\right] \approx [10.30, 47.78]$
- **為何偏好小變異性？** 較小的變異性代表四分衛的表現穩定度高，這讓教練在戰術規劃時有更高的可預測

c. 每傳碼數變異數 σ^2 與標準差 σ 的 95% 置信區間

- **變異數區間** (σ^2): $\left[\frac{15 \times 6.00}{27.488}, \frac{15 \times 6.00}{6.262}\right] \approx [3.27, 14.37]$

  $(df = 15)$

- **標準差區間** (σ): 對上述結果開根號，約為 $[1.81, 3.79]$。

d. 檢定標準差是否不同於 $\sigma = 4$ ($\alpha = .05$)

1. **建立假設**: $H_0 : \sigma^2 = 16$ vs $H_a : \sigma^2 \neq 16$
2. **檢定統計量**: $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{15 \times 6.00}{16} = 5.625$
3. **判定**:
   - 雙尾檢定拒絕域： $\chi^2 > 27.488$ 或 $\chi^2 < 6.262$
   - 因為 $5.625 < 6.262$，**落入拒絕域**。
4. **結論**: 有顯著證據顯示該四分衛每傳碼數的標準差與 4 不同 (實際上較小，表現更穩定)。

13. Instrument Precision A precision instrument is guaranteed to read accurately to within 2 units. A sample of four instrument readings on the same object yielded the measurements 353, 351, 351, and 355.

a. Test the null hypothesis that $\sigma = .7$ against the alternative $\sigma > .7$. Use $\alpha = .05$.

b. Find a 90% confidence interval for the population variance.

13. 儀器精度檢定 (卡方檢定)

數據： $\{353, 351, 351, 355\}$

- 平均值 $\bar{x} = 352.5$
- 樣本變異數 $s^2 = 3.667$

a. 檢定 $H_0 : \sigma = .7$ vs $H_a : \sigma > .7$ ($\alpha = .05$)

1. **統計量**: $\chi^2 = \frac{(4-1) \times 3.667}{0.7^2} = \frac{11}{0.49} \approx 22.45$
2. **臨界值**: $df = 3, \chi^2_{.05,3} = 7.815$
3. **結論**: $22.45 > 7.815$，**拒絕** H_0。儀器的標準差顯著大於 0.7，精確度不如預期。

b. 90% 置信區間

- 查表： $\chi^2_{.05,3} = 7.815$， $\chi^2_{.95,3} = 0.352$
- 區間： $\left[\frac{3 \times 3.667}{7.815}, \frac{3 \times 3.667}{0.352}\right] \approx [\mathbf{1.41}, \mathbf{31.25}]$